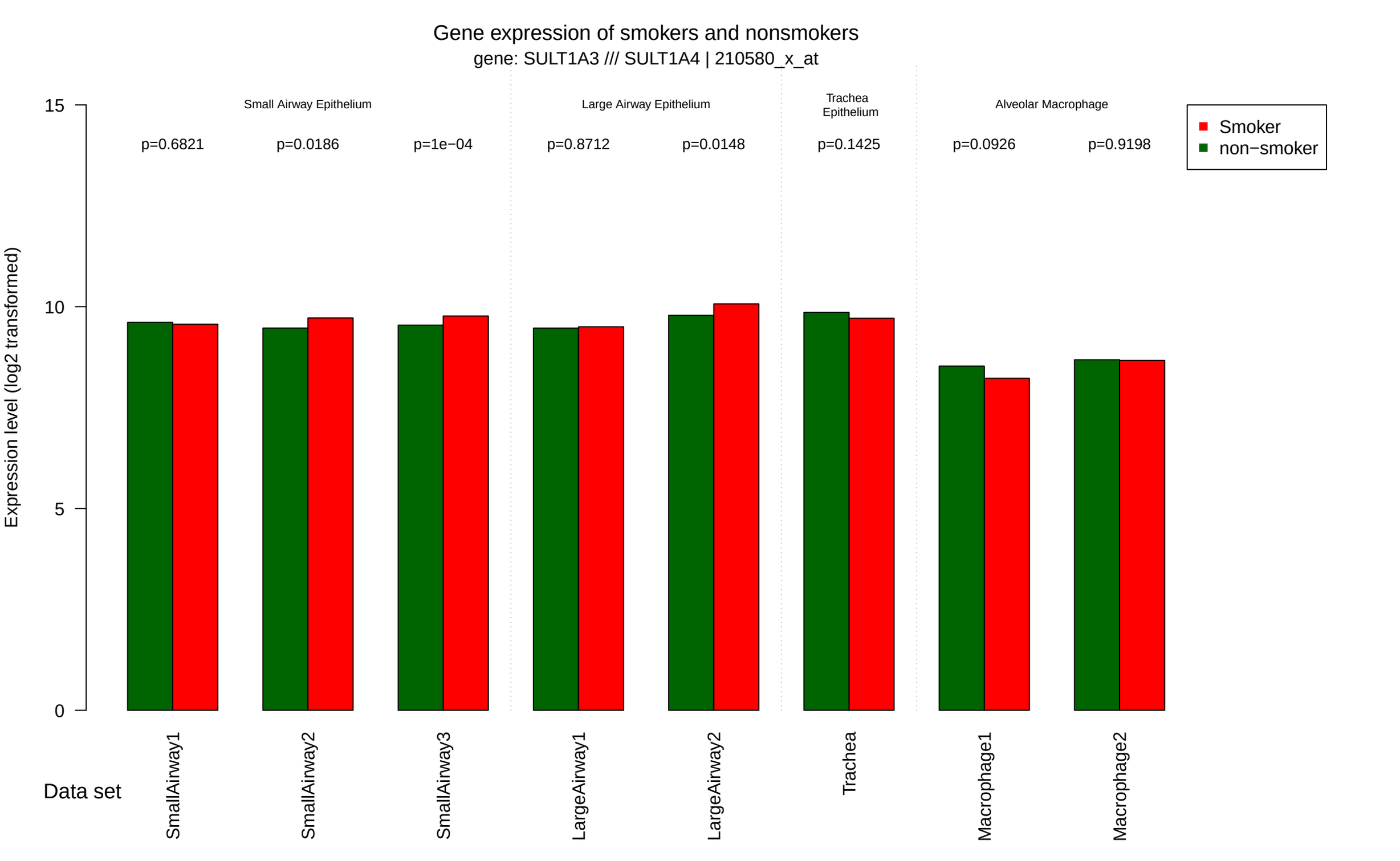
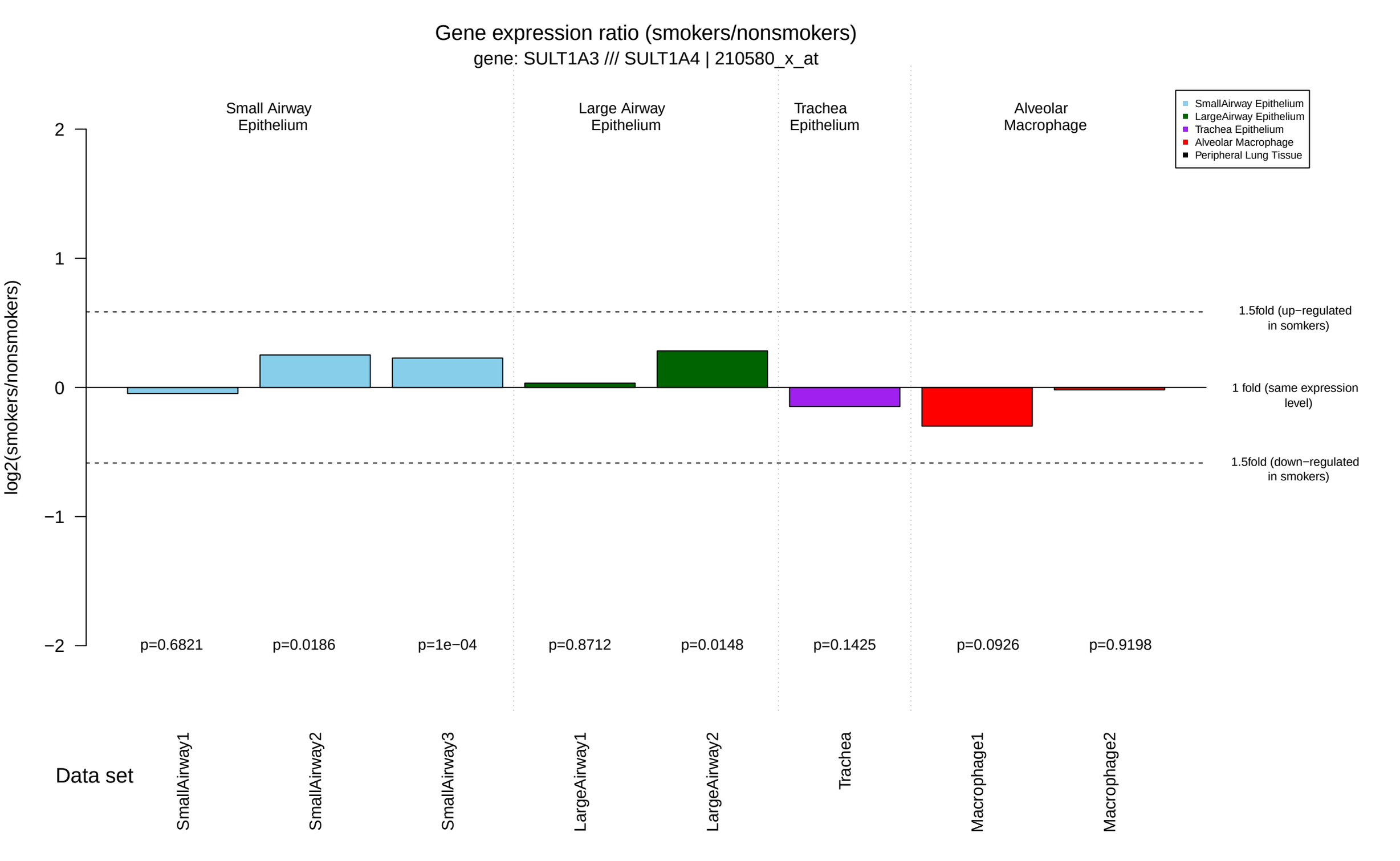
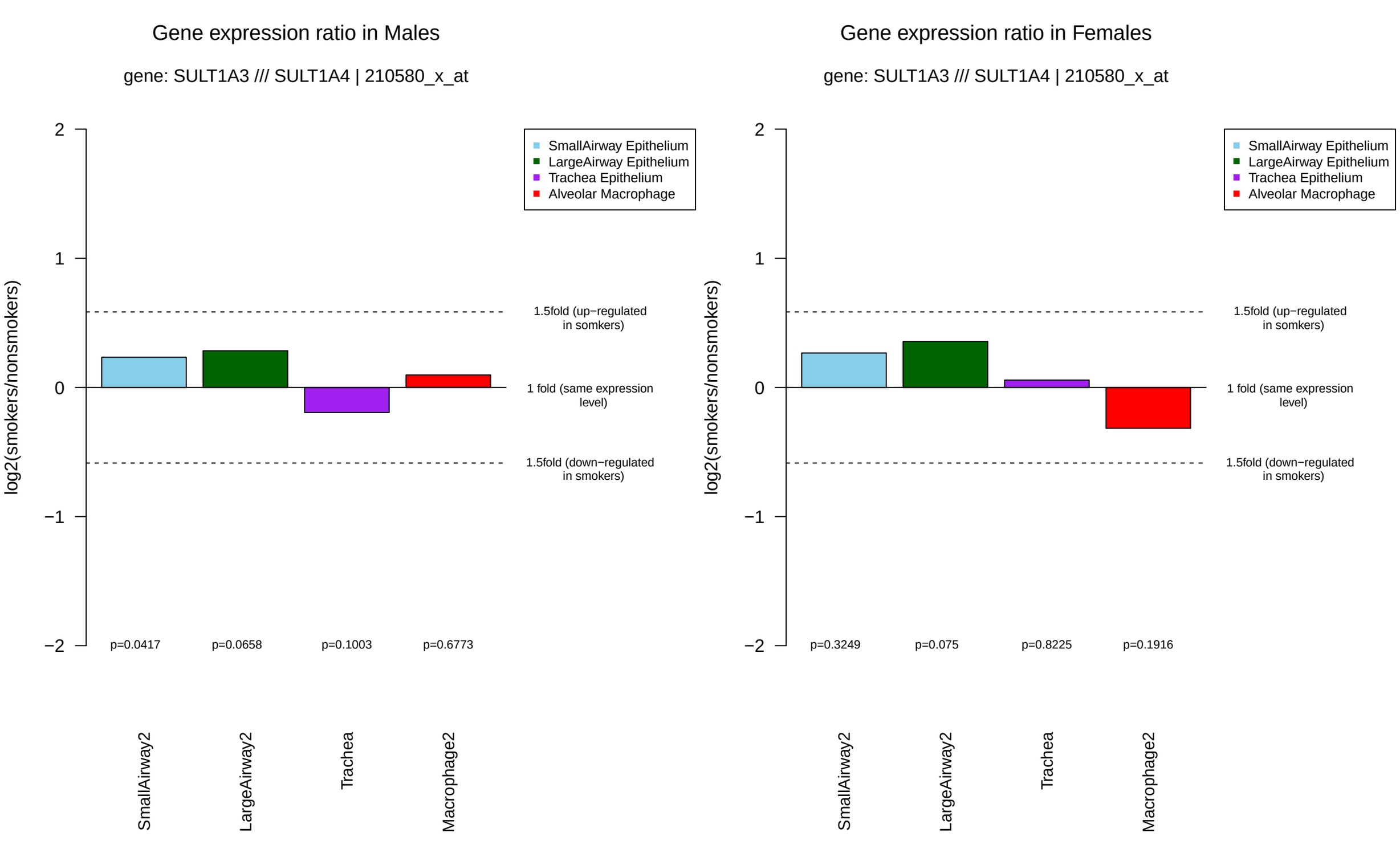
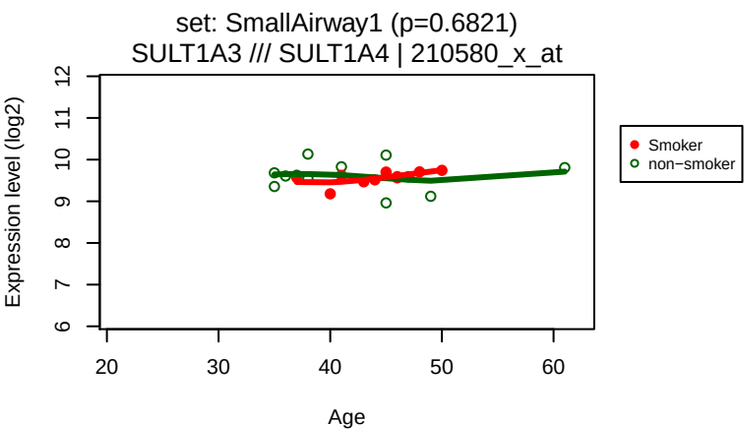
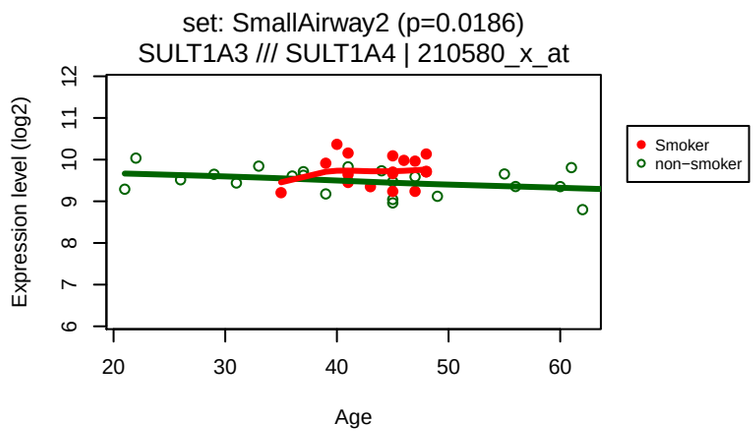
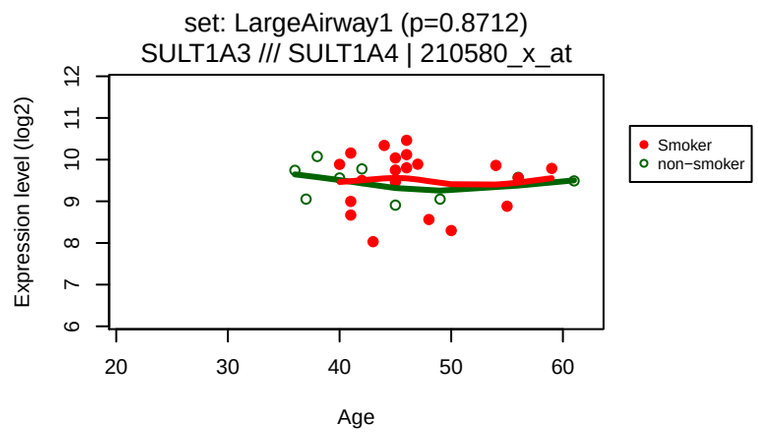

Gene expression of smokers and nonsmokers
gene: SULT1A3 /// SULT1A4 | 210580_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6821
p=0.0186
p=1e−04
p=0.8712
p=0.0148
p=0.1425
p=0.0926
p=0.9198
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

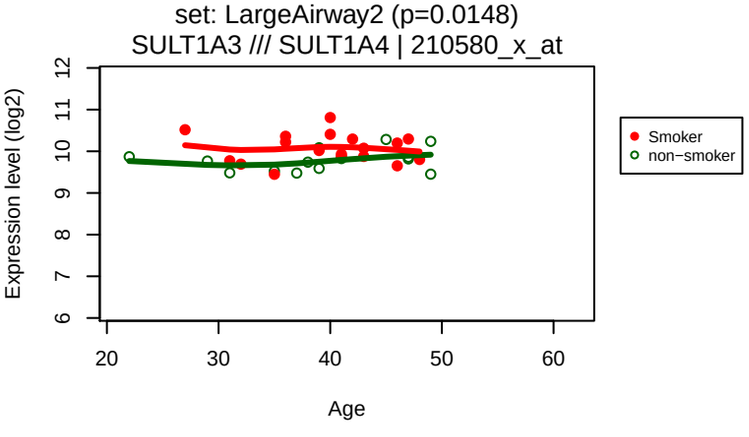

Gene expression ratio (smokers/nonsmokers)
gene: SULT1A3 /// SULT1A4 | 210580_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6821
p=0.0186
p=1e−04
p=0.8712
p=0.0148
p=0.1425
p=0.0926
p=0.9198
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

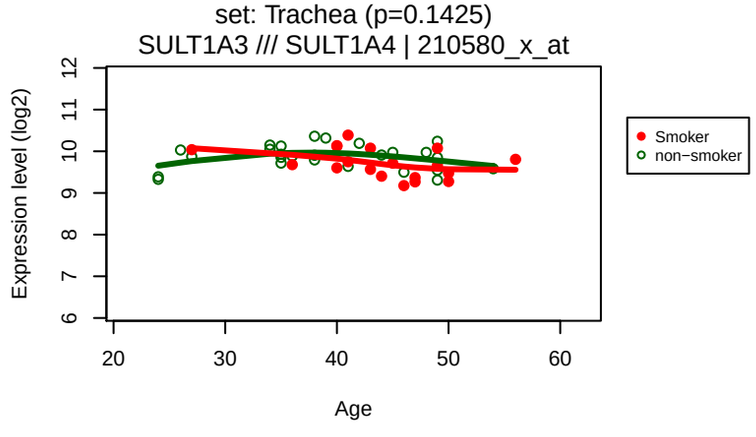

Gene expression ratio in Males
gene: SULT1A3 /// SULT1A4 | 210580_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.0658
p=0.1003
p=0.6773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SULT1A3 /// SULT1A4 | 210580_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3249
p=0.075
p=0.8225
p=0.1916
SmallAirway2
LargeAirway2
Trachea
Macrophage2

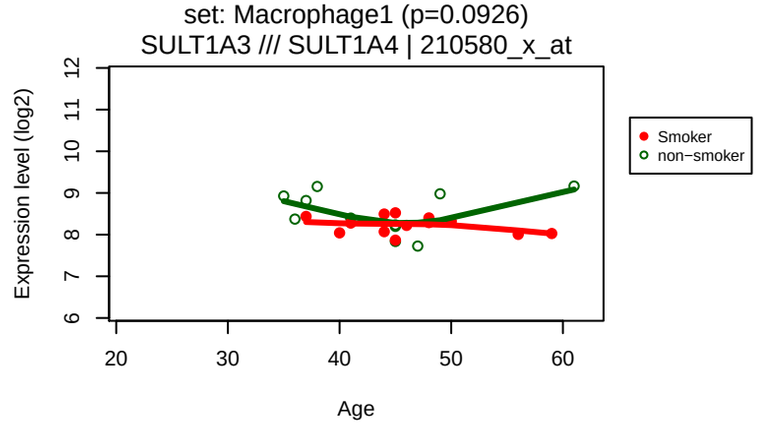

set: SmallAirway1 (p=0.6821)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker

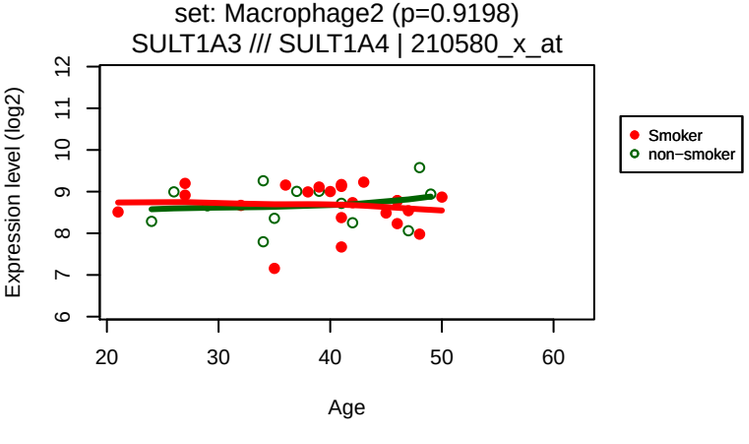

set: SmallAirway2 (p=0.0186)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker

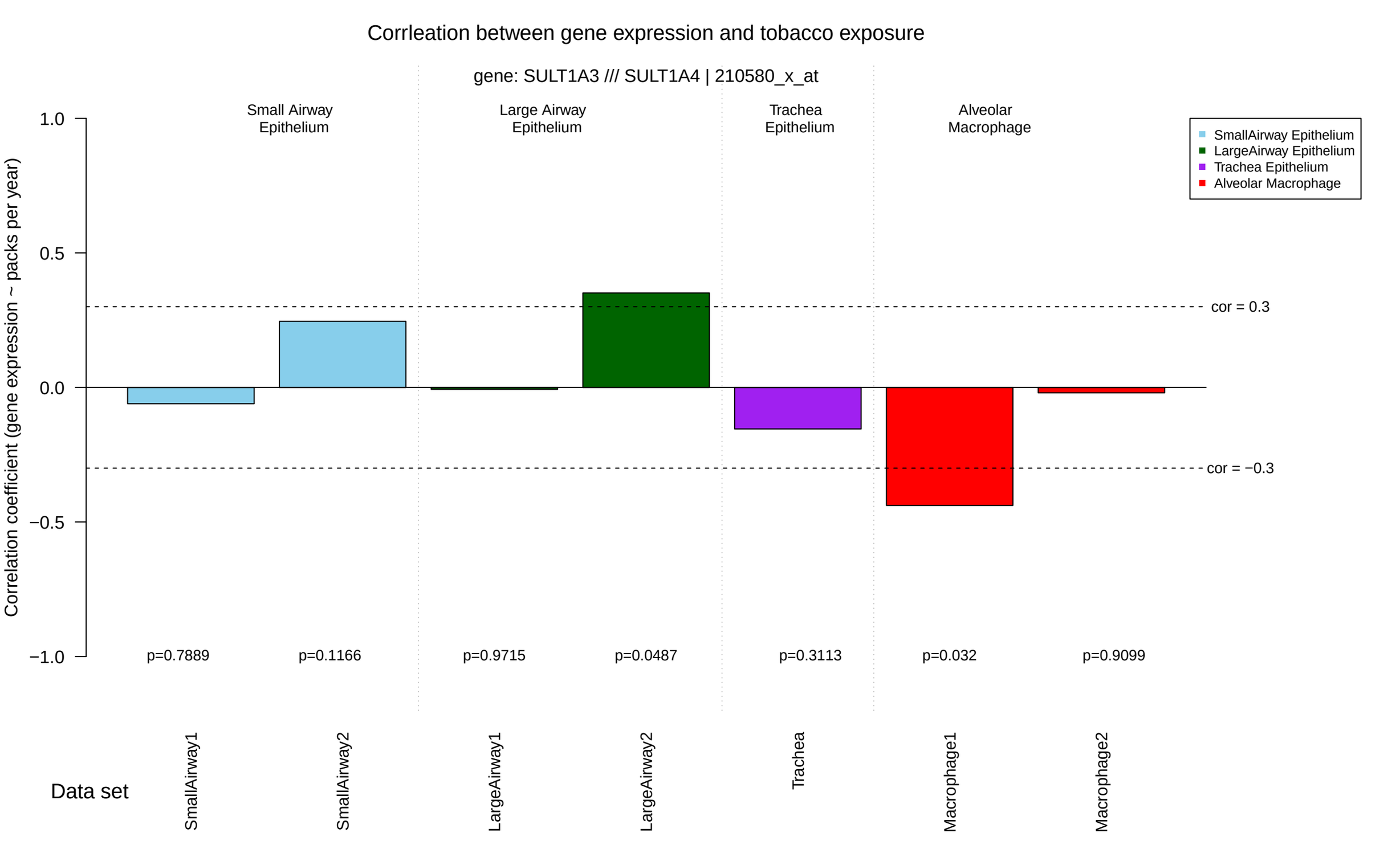

set: LargeAirway1 (p=0.8712)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0148)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1425)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0926)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9198)
SULT1A3 /// SULT1A4 | 210580_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SULT1A3 /// SULT1A4 | 210580_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7889
p=0.1166
p=0.9715
p=0.0487
p=0.3113
p=0.032
p=0.9099
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2